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SRN: PES2UG24CS350

Subject: Software Engineering

Section: F

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Step 2: Running the Original Script

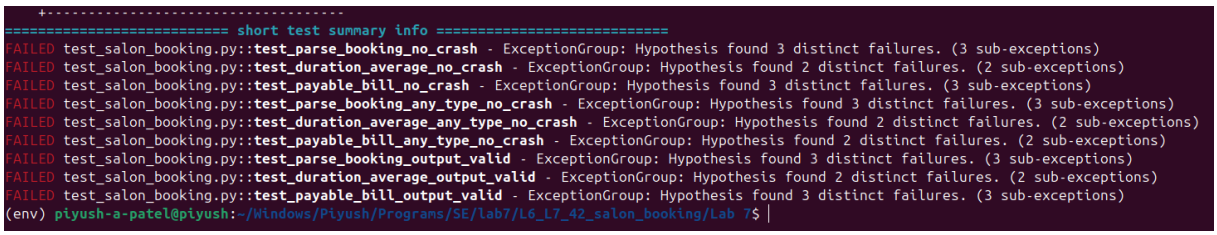
```
(env) piyush-a-patel@piyush:~/Windows/Piyush/Programs/SE/lab7/L6_L7_42_salon_boo
king/Lab 7$ python salon_booking.py
==== Running parse_booking ====
Enter service:duration in min (e.g. haircut:45): haircut:45
Service and Duration: ('Haircut', 45.0)

==== Running duration_average ====
Enter durations as CSV (e.g. 30,45,60): 30,45,60
Average Duration: 45

==== Running payable_bill ====
Enter bill|discount (e.g. 800|100): 800|100
Payable Bill: 844.0
```

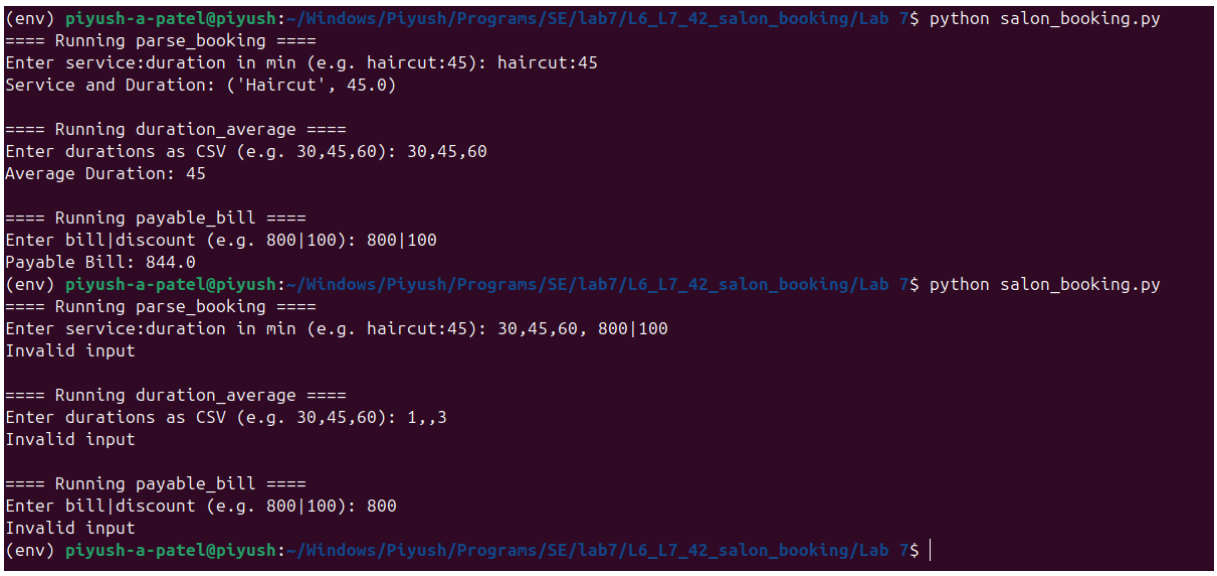
Observation: The original script runs correctly when given well-formed input such as haircut:45, 30,45,60 and 800|100. It performs no input validation, so malformed input would crash it.

Step 4: Fuzz Tests on the Original Script



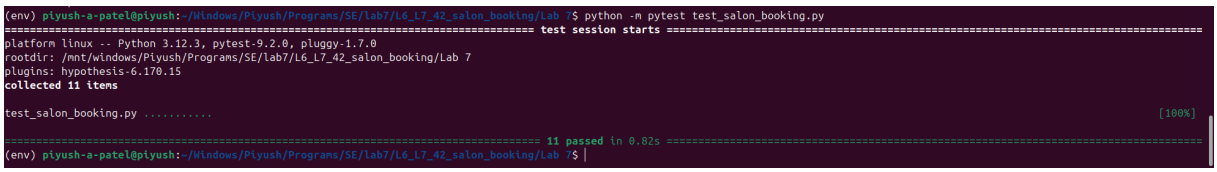
Observation: 9 of the 11 tests failed. Hypothesis found crashing inputs in all three functions, such as None, a missing ':' or '|', empty CSV items, non-numeric values and nan/inf.

Step 5: Running the Fixed Script



Observation: Valid input gives the same output as before. Invalid input such as 30,45,60, 800|100, 1,,3 and 800 now prints Invalid input instead of raising an exception.

Step 6: Fuzz Tests on the Fixed Script



Observation: All 11 tests passed against the fixed script. Hypothesis could no longer find any input that crashes the three functions or produces an invalid result.